

The Role of Artificial Intelligence in Idea Management Systems and Innovation Processes: An Integrative Review

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ABSTRACT

The role of artificial intelligence (AI) in idea management systems (IMS) and innovation processes has recently been a topic of significant research interest. AI has been acknowledged for its potential to enhance innovation activities by providing support in various aspects. The intersection of these areas offers promising opportunities for improved idea generation, classification, development, and evaluation. However, AI's impact is not equally present for the different stages of an innovation process, showing more prominence in the idea generation stage. Through an integrative review, we can explore how AI has contributed to different steps of innovation processes implemented through IMS. AI-driven approaches, so far, have been opening possibilities to manage creative processes, such as automating specific tasks, analyzing large amounts of data to identify patterns and trends, and providing real-time feedback to enhance ideation and decision-making. Assessing the contribution of AI to innovation and creativity is vital in understanding its potential influence on the future developments of IMS and innovation processes.

CCS CONCEPTS

• Artificial Intelligence; • Data Management Systems; • Machine Learning;

KEYWORDS

Idea Management Systems, Innovation Process, Idea generation

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1 INTRODUCTION

Recently, idea management system (IMS) providers have been integrating artificial intelligence (AI) technologies like machine learning (ML), deep learning, and natural language processing to power up their systems' functionality. AI can be defined as the field of

computer science focusing on developing systems capable of emulating human intelligence, which involves the interpretation of data by machines, learning from it, and applying the knowledge to accomplish certain tasks autonomously [1]. On the other hand, IMS are formalized systems within an organization that facilitate the process of idea generation, evaluation, and more by following a structured and controlled process approach [2]. How AI has the potential to, in fact, revolutionize IMS is shown by [3]. Organizations that implement IMS are of all sizes and industries. IMS are mostly tailored according to the organization's needs and based on the innovation strategy in place. Each IMS follows a structured step-by-step process that facilitates the flow of ideas from the very early stages to the execution stage, which, in most cases, is related to new product development with an end goal. Depending on the type of IMS, the passive, active, or mixed ones [4]. IMS can support an entity's open innovation practices, and it belongs to the innovation management activities, as well as among the knowledge creation and acquisition tools [5]. Conversely, AI can come in handy to automate tasks, analyze big data to create (new) insight and improve decision-making. Integrating AI and IMS can boost innovation capacity [6], motivate employee-driven ideas [7], and build the entity's competitive advantages in the market [8]. IMS has become the strongest catalyst for the sales growth of new products by 7.2 % [9].

Research gaps in the intersection of AI and IMS present significant opportunities for further exploration. Some of the research questions proposed in the body of literature so far are: How can AI enhance innovation management and new product development [10] research not only on commercially available web-based IMS but also private and non-commercial IMS, like the ones for educational purposes [4], and how do the stages of a firm's innovation process shape firm's AI adoption for innovation [11], to name a few. [12] referred to the process of generating innovations using computers and databases of information, experiences, and knowledge as the "technocratic route to innovation." Based on the author's experience and observations, AI has already been implemented in a handful of IMS; however, from the literature review analysis, the value added of AI specifically or its potential is not captured in research. This study aimed to conduct an integrative review to examine the role of AI in IMS so far, including the AI implementation in different steps of innovation processes and the types of AI present.

2 METHODOLOGY

As [13] described, an integrative review is the "broadest type of research review and provides a framework to investigate complex concepts. This allows for literature from a diverse range of methodologies, and it is intended to address new and emerging topics that vary long from old to new research, using multiple databases to

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identify relevant literature from business, innovation, technology, engineering, and other related fields [14].

Before formalizing a search strategy, an initial scoping search was completed to ensure the correct and relevant terms were used. Research has captured the AI leveraged in innovation management. However, IMS are outlined as a more defined term than idea management, which belongs to innovation management. The keyword search included the following terms: ("artificial Intelligence" OR "machine learning") AND "idea management system." There was no time limit in the scope of our review. Electronic database searches were performed in Scopus, IEEE Explore, Engineering Village, and Google Scholar to get an overview of the prior literature. The selection of the databases should ensure that we base the integrative review on authoritative outlets. The initial search resulted in 179 studies. This is complemented by reference tracking and citation searching, adding 28 results here. Duplicates were found and removed. The publications' titles, abstracts, and conclusions were initially analyzed to determine if they focus on IMS, AI, and/or ML. Furthermore, only research in English was included. The second most popular language among studies was Spanish. Another inclusion criterion was related to the content of the materials, provided they are related to the area of innovation and show the inclusion of different types of IMS. A few research results used "collective intelligence" instead of our "artificial intelligence" keyword. IMS falls under the label of collective intelligence tools [15], but with no specifications if AI is implemented or not.

After these steps- identification of the research, initial screening, and sifting through the eligibility criteria- 97 publications were used as the sample. The authors took one more step of looking closer at the materials for specific connections between AI and IMS, and the final number of studies included in the data analysis is 59. Data abstraction and retrieval involved thoroughly reading the articles to ensure the notion of AI and IMS was appropriately captured. The data was organized in a table format, sorted by publication date, from the oldest study (1983) to the most recent one (2024), followed by quoting or summarizing all the important sections related to AI and IMS. The author conducted a content analysis of the data, identifying categories, common patterns, themes or topics, and relationships between the terms. This process identified three key categories: (1) AI in certain innovation steps of an innovation process in IMS, (2) types of AI algorithms implemented, and (3) potential use cases of AI in IMS.

3 RESULTS

Studies identified by the review process and included in the data analysis covered a variety of studies: 27 journals, 17 reports (like Theses, lecture notes, posters, or newsletters), 8 book chapters, and 7 conference proceedings. Furthermore, studies included in the review come from a wide array of journals spanning different disciplines. This variety promotes a multidisciplinary approach to knowledge synthesis, strengthening the integrative review's validity and comprehensiveness. As mentioned before, the results of the study are the following:

3.1 Artificial intelligence across an innovation process

The definition of AI within the context of innovation involves its ability to simulate human intelligence, and its applications are in different industries. AI stimulates organizational growth and development in business innovation through intelligent systems, such as knowledge management systems [16, 17]. The studies capturing AI and IMS outline a wide variety of innovation processes, mostly tailored according to different innovation strategies that businesses have in place and depending on their availability of resources for innovation.

To capture the use case of AI in the different steps of innovation processes, the author uses the Creative Idea Solution (CIS) model, developed in 2000 at the Centre for Ideas and Innovation at the Danish Technological Institute [18]. Considering how AI is seen as a tool to handle problems of information and data overflow, a model like CIS, which builds on the chaos theory, uses a systematic step-by-step process to overcome chaos, in this case, the immense amount of data that idea creators need to review. On the other hand, as [5] classifies IMS as knowledge management tools, and we established that IMS implements innovation processes, the model of choice, CIS, is based on the notion of having a constant supply of new knowledge on each step of the innovation process. Furthermore, CIS aims to empower the creator of an idea. The new knowledge and insight provided in the process raise their chances of influencing the idea development and getting a sense of ownership in developing new products. Such ownership fuels the idea creator's motivation to improve ideas, no matter the timeliness of the feedback loops that the business might have implemented [19].

To further explain the steps of CIS model, let's start with the lateral focus which is to define and align the idea creators with the given task, case, problem, or opportunity at hand. The PRE-ject phase corresponds with the concept of front-end innovation, where input creation represents ideation, or, differently, idea generation, and idea treatment is idea evaluation. The PRO-ject phase would correspond with the New Product development activities and the POST-ject with the commercialization of the product. The CIS process is implemented in the IMS called Rosetta, which the author now uses for educational purposes in engineering and innovation courses [20]. In Rosetta, the inputs and ideas of PRE-ject phase are colour-coded, intending to categorize the ideas as incremental, disruptive, radical and ambivalent. We can describe this as the idea classification step. The idea selection or idea filtering step happens based on action-convergence due to the consistent supply of new knowledge and activities to develop further the ideas, and not based on decision-convergence, meaning postponing judgment on ideas to the next iteration of the CIS process [20].

The following captures each innovation process step using the CIS model and lists the relevant studies. With regard to idea generation, AI enhances this step through the connection with social network data and online communities [21–25]; knowledge discovery with data mining [26]; and the use of AI-supported idea generation tools [8, 27, 28] like a recommender system [29] or chatbots [30, 31] and trends analysis [32].

Idea classification has been introduced within the idea selection step [33], and it comes in handy to support the innovation teams,

more specifically, the facilitators of the innovation process [15]. Annotations have been one approach for idea classification [24, 34–38] or through comparing and analyzing the idea content for similarities [29, 34, 37, 40–44]. Idea selection, on the other hand, appears to be empowered by the potential that Big Data provides to help the decision-making process of the idea creators and evaluators [41, 45]. Annotations are proposed here, too, as a tool to help with the efficiency of selecting ideas [36, 42, 46].

The use of AI tools for idea selection seems to help reduce the biases of the human actor [47], even though the human actor is mandatory for the annotation part [36, 48], which is later expanded with the AI's recommendations.

3.2 Artificial Intelligence Algorithms in Idea Management Systems

Various methodologies emerge in exploring the landscape of AI tools within IMS, reflecting the multifaceted nature of innovation processes. Drawing insights from [49, 50], the utilization of AI in IMS unveils a spectrum of approaches to foster creativity and enhance idea generation. AI tools play a pivotal role in shaping the ideation process, from synthesizing novel combinations of familiar ideas to exploring conceptual spaces and facilitating transformations for previously unattainable ideas. This part of the review results delves into the implications of different AI applications, as highlighted by [50], shedding light on the transformative potential of AI in driving innovation and knowledge creation within organizational settings.

ML is the most popular algorithmic tool for IMS, followed by semantic web applications. Support Vector Machine is mainly used for idea classification [51, 52]. In addition, Natural Language Processing (NLP) is implemented to analyze idea text content for many ideas in IMS [38, 53, 54]. However, [38] recommends going beyond NLP, as their shallow comprehension of natural language constrains them. The limitations of these techniques stem from the inherent complexity of human language used in idea description, which involves nuances, context, and ambiguity that are challenging for algorithms to grasp fully. [55] introduced a comprehensive decision-making framework that leverages neural networks and fuzzy logic to enhance decision-making efficiency. Their focus was optimizing the evaluation process, particularly in the initial phases of idea development.

Topic modeling is growing in popularity with regards to providing a structured approach to analyze and categorize large volumes of unstructured text data among idea descriptions, enabling the identification of patterns, trends, and themes within the idea repositories [41]. These models facilitate automatic clustering and organizing of ideas based on semantic similarities, allowing for efficient idea exploration and retrieval [32]. In addition, TF-IDF has been leveraged to handle the problem of information overload in the context of open innovation, proposed as a novel method for valuable idea identification through the case of an IMS, organizing data from customers in an online community [56].

Lastly, when utilized as an AI tool in idea management, the semantic web offers significant advantages in structuring and enhancing the innovation process. Semantic web mining techniques allow for efficient management of ideas [8, 57], facilitating the extraction of

meaningful information and providing insight for idea generation [58], description [2, 594], classification [29, 35, 36], selection or filtering [60], development [29], evaluation [39, 61].

3.3 Artificial intelligence and its potential in the future of Idea Management Systems

How AI can empower idea management was captured by studies as soon as 1983 by [62], proposing that it has the capacity to mimic human cognitive functions by effectively retaining and retrieving ideas through a process known as 'inference.' [63] has hypothesized that AI methodologies can facilitate the generation of innovative ideas through three primary mechanisms: by synthesizing fresh blends of existing concepts, by delving into the possibilities within conceptual frameworks, and by effecting transformations that pave the way for the emergence of ideas that were previously inconceivable. In addition, the potential of big data in idea selection has yet to be captured and explored [45]. AI is evaluated to have the capacity to reshape the innovation process and IMS there, following [64] recommendation on the "Sense-Think-and-Act process," which can have an intelligent agent through all the steps, performing tools like deep learning algorithms. Also, [65] proposed using AI applications to identify lead users by empowering the approaches, i.e., screening, pyramiding, social networks, ethnography, and even combinations of those. [24] highlights the premise of utilizing AI-driven semantic clustering of idea content as especially beneficial for nurturing original contributions by drawing inspiration from extensive repositories of past ideas (i.e., MyStarbucksIdea.com). AI-based support of idea management can be as niche as to support organizing their thoughts into potential future startup ideas [66], by actively collecting these thought impulses, proactively structuring them, further processing, and bring them back to the entrepreneur's attention later.

4 DISCUSSION

The above-synthesized literature offers insight into ML, big data, and semantic web applications, to mention a few, in optimizing IMS and assisting with the information overflow in the innovation processes. Studies have been specific in building idea ontology for knowledge management and to increase data awareness for exploiting further idea annotations [35, 39, 60]. The review shows that studies supporting the use of AI in idea generation seem more popular compared to the work of AI in other steps of an innovation process implemented in IMS. Neglecting the idea generation step would limit the pool of innovative concepts that are available for further development and realization. However, the author proposes to research AI implementation in other steps further to strike a balanced approach that includes idea development and idea evaluation. Rumor has it that there is not much need for ideas as there are plenty and that the challenge lies in determining whether an idea is merely good or a truly viable business opportunity [67]. In this light, the author is extending the work in this area to use AI to develop radical innovation intentionally [20].

Through the review, it has come to attention, though not repentantly, that AI tools are applied in IMS to improve teamwork and collaboration online [6, 31, 33, 68, 69]. Furthermore, emphasis is given to the importance of fostering a culture of innovation across

the organization to ensure that the implementation of AI tools in IMS is productive [44]. The author proposes to further the work on the hypothesis that if the integration of AI tools happens after the companies have gone through a digital transformation journey, the chances are that there will be more acceptance across the idea creators to onboard and use the IMS. This is to be accompanied by a leadership (middle-management team supporting exploitations and exploration, meaning the organization is more ambidextrous [70].

5 CONCLUSION

This paper presents an integrative review of the role of AI in IMS and innovation processes. The review used 59 studies for data analysis, which led to the identification of the results in three key categories: (1) AI in certain innovation steps of an innovation process in IMS, (2) types of AI algorithms implemented, and (3) potential use cases of AI in IMS. The study's result focuses on the utilization of AI across various stages of the innovation process, using the CIS model as a framework, which relies on the constant supply of new knowledge and insights to empower idea creators. AI's role involves simulating human intelligence, particularly in idea generation, classification, development, and evaluation. Within IMS, AI tools enhance different steps through social network data analysis, online community engagement, data mining, recommender systems, chatbots, and trend analysis. Idea classification and selections benefit from AI-powered tools for annotation, decision-making, and analysis by using methodologies like NLP, semantic web, and topic modeling, to mention a few. The results also explore the transformative potential of AI algorithms in IMS, including their role in synthesizing novel (radical) ideas, facilitating team collaboration, and impacting the organizational culture. The study suggests future research directions, such as investigating AI's impact on other stages of innovation (besides the focus on idea generation) and exploring AI's integration in IMS post-digital transformation processes to enhance acceptance and effectiveness within organizations. In conclusion, AI has the potential to enhance IMS and innovation processes. AI can streamline and improve the ideation and decision-making processes by automating tasks, analyzing data, providing real-time feedback, and facilitating collaboration.

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